

Schedule-based analysis of airborne transmission risk in public transportation systems

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ABSTRACT

Airborne diseases raise the question of transmission risk in public transportation systems. However, quantitative analysis of the effectiveness of transmission risk mitigation methods in public transportation is lacking. The paper develops an airborne transmission risk modeling framework based on the Wells-Riley model using as inputs transit operating characteristics, schedule, Origin-Destination (OD) demand, and virus characteristics. The model is sensitive to various factors that operators can control, and external factors that may be subject of broader policy decisions. The model is utilized to assess transmission risk as a function of OD flows, planned operations, and factors such as mask-wearing, ventilation, and infection rates. Using actual OD and AVL data from the Massachusetts Bay Transportation Authority (MBTA) Red Line, the paper explores the airborne transmission risk under different infection rate scenarios. The paper assesses the combined impact from viral load related factors and passenger load factors. Increasing frequency can mitigate risk, but cannot fully compensate for increases in infection rates. Imbalanced passenger distribution on different cars of a train increases the overall system-wide infection probability. Spatial infection rate patterns should also be considered during policymaking. For lines with branches, demand distribution among the branches is important and headway allocation adjustment can reduce risk.

Introduction

Major pandemics, such as COVID-19, has far-reaching societal impacts (Coronaviridae Study Group of the International Committee on Taxonomy of Viruses, 2020). Public officials have taken various measures to control the spread of the disease, such as closing non-essential businesses and facilities, encouraging non-essential workers to stay home, and mandating mask wearing (Humphreys, 2020). Public transportation and its role in the spread of infectious diseases has been a significant part of the discussion.

Various studies in the literature have examined the impact of pandemics on public transportation ridership, attitudes towards using public transportation, etc. As a result, and in combination with local directives and business shut-down mandates, a dramatic decrease in transit ridership occurred in all metropolitan areas at the early stages of the latest pandemic such as COVID-19 (Dickens, 2020; Jenelius and Cebeacauer, 2020; Liu et al., 2020; Orro et al., 2020; Park, 2020).

Researchers have also studied the role of public transportation systems in the transmission of airborne diseases, and found correlations

between infectious diseases including influenza-like or COVID-19 illnesses with public transportation systems (Goscé and Johansson, 2018; Harris, 2020). Researchers also analyzed various cases of COVID-19 transmission in bus trips to directly assess the risk of transmission in public transportation (Luo et al., 2020; Shen et al., 2020). Besides transmission risk in the buses, Shinohara et al. (2021) evaluated the transmission risk on commuter trains and found that the transmission risk can be significantly reduced with ventilation. Zhu et al. (2021) studied various travel modes and travel destinations and the impacts on COVID-19 transmission in different cities and found that different travel modes have impacts on the number of cases in the cities.

Several studies applied and adapted the concept of contact networks and the “susceptible-infectious-recovered” (SIR) framework for transmission modeling infectious diseases (Chen et al., 2020; Prem et al., 2020). Mo et al. (2021) developed a public transit contact network and found that partial closure of bus routes cannot fully contain the spreading of an epidemic. The study by Hanig et al. (2023) examines COVID-19 precautions in public transit, focusing on risk reduction and cost trade-offs. Using a Monte Carlo Analysis, the researchers assessed

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strategies like masks, additional buses, ride-hailing, and autonomous vehicles (AVs). The findings suggest that AVs may be cost-effective alternatives to crowded buses during the pandemic. These studies focus on the dynamic evolution of transmission risk in the general population of a large metropolitan area, using a time horizon of days or months. They are dynamic, modeling the evolution of the spread of the disease over time but do not provide detailed accounting of how everyday operations of public transportation systems contribute to the transmission, and are not sensitive to the operational characteristics of transit services.

Other studies aim to assess the transmission risks in small, confined spaces in relatively short time periods, normally within a few hours. For such cases, Riley et al. (1978) proposed a model and used it for the epidemiological study of a measles outbreak. The model, known as the Wells-Riley model, has been used extensively by researchers to study the transmission risk of various respiratory diseases, including influenza, SARS, etc. inside hospital wards, classrooms, and transit vehicles (Andrews et al., 2013; Barnett, 2020; Chen et al., 2011; Dai and Zhao, 2020; Fennelly and Nardell, 1998; Furuya, 2007; Ko et al., 2004; Rudnick and Milton, 2003; Stephens, 2012). The original model estimates the expected number of infections as a function of the ventilation rate, exposure time, number of carriers, and the infectiousness of the disease. The studies that looked at transmission risk in transit vehicles treated public transportation vehicles as a close-to-static indoor environment, assuming a fixed number of people and infectors (Andrews et al., 2013; Barnett, 2020; Chen et al., 2011; Dai and Zhao, 2020; Fennelly and Nardell, 1998; Furuya, 2007; Ko et al., 2004; Rudnick and Milton, 2003; Stephens, 2012). The exposure time is also assumed to be constant. As a result, they are more appropriate for intercity travel and do not capture the temporal and spatial characteristics of public transportation demand in urban areas. One exception is the study by Zhao et al., 2022, which adapted the Wells-Riley model to consider the mask wearing behavior and vaccination percentage and to explore the COVID-19 transmission risk for a particular bus route in Houston, Texas. However, their study also considers a fixed number of people and infectors, neglecting the dynamic and spatio-temporal characteristics of public transportation.

Other than the Wells-Riley model, computational fluid dynamics (CFD) models (Foster and Kinzel, 2021; Vignolo et al., 2023) are also used for estimation of transmission risk of COVID-19. The Wells-Riley model, which assumes a well-mixed air environment, has been extensively used due to its simplicity and ease of interpretation. However, this assumption of uniform mixing is a notable limitation, as it fails to account for the complexities of real-world airflow and particle distribution. To fill this gap, CFD models offer a more sophisticated analysis by capturing the intricate dynamics of airflow, temperature gradients, and localized particle dispersion (Mohamadi and Fazeli, 2022). Several studies have conducted comparative analyses between the Wells-Riley and CFD models. The analytical Wells-Riley approach demonstrated relatively good performance overall, but inhomogeneities in airflow were observed to cause significant underprediction of risk in some cases, especially at high ventilation rates (Foster and Kinzel, 2021; Vignolo et al., 2023). The Wells-Riley model produced relatively accurate results in scenarios with low ventilation, with errors as low as 6 %. However, in high-ventilation environments, the model tended to underpredict transmission risk (around 29 %). To address the limitations from the Wells-Riley model, some studies have explored combining the Wells-Riley model with CFD simulations to enhance the accuracy of transmission risk assessments (Wang et al., 2022).

Despite limitations, the Wells-Riley model continues to be widely used in academic research due to its simplicity. To further apply the model to transportation operations, Zhou and Koutsopoulos (2021) proposed a modified Wells-Riley model for risk analysis in public transportation systems that captures passenger flow spatio-temporal characteristics and actual operations. The model is utilized to assess overall risk as a function of passenger flows, actual operations, and factors such as mask-wearing, ventilation, infectiousness of disease, and infection rate in the population. The model is integrated with a

microscopic simulation model of subway operations (SimMETRO) (Koutsopoulos and Wang, 2007; Zhou et al., 2020) for train trip-level analysis of transmission risk. While the model is sensitive to the various factors that impact risk, its reliance on a detailed simulation model for trip-based analysis limits its applicability.

The objective of this study is to present a simplified model, analogous to a planning tool, that can quickly analyze and quantify transmission risk under various operating strategies and other risk mitigation approaches so that decision makers can make informed decisions. It is worth noticing that viruses can be transmitted through two primary routes: fomite (surface contact) and aerosol (airborne particles). Respiratory viruses such as SARS-CoV-2 are thought to be transmitted through aerosol (Greenhalgh et al., 2021), typically during prolonged close contact, but there is a possibility and potential for fomite transmission of SARS-CoV-2 (Short and Cowling, 2023). Even the COVID-19 predominantly spreads via aerosols, the fomite route remains important for other infectious diseases. This study specifically focuses on aerosol transmission, reflecting its relevance in the current pandemic context.

The main contribution of this study is the development and application of a quantitative framework for risk assessment that is sensitive to demand and service delivery in public transportation systems. It proposes a method to model the risk of transmission of airborne diseases based on the Wells-Riley model, and evaluates the transmission risks under planned service conditions, as opposed to trip-based conditions, using as inputs Automatic Fare Collection (AFC) and Automatic Vehicle Location (AVL) data. The framework can be utilized to evaluate the effectiveness of agency-controlled measures such as operating strategies and ventilation, as well as exogenous factors such as mask-wearing, infection rates in the population, and infectiousness of the disease. It is sensitive to the spatial characteristics of demand and infection rates, and applicable to any transit system.

The paper is organized as follows. Section 2 discusses the derivation of a Wells-Riley based model incorporating the demand and supply dynamics in a public transportation system in a general analytical framework. Section 3 introduces a case study using data from the Massachusetts Bay Transportation Authority (MBTA) Red Line. Section 4 discusses the results from the case study and evaluates the effectiveness of different mitigation strategies. Section 5 concludes the paper.

Methodology

The Wells-Riley model

Riley et al. (1978) proposed the Wells-Riley model for infection probability evaluation in indoor environments, considering the intake dose of airborne pathogens in terms of the number of infectious particles.

$$P = 1 - \exp\left(-\frac{I p q t}{Q}\right) \quad (1)$$

where P is the probability of infection, I the number of infectors (carriers), p the breathing rate per person (m^3/hour), q the quanta generation rate (quanta/hour), t the exposure time (hour), Q the room ventilation rate of clean air (m^3/hour).

The Wells-Riley model is adopted in the World Health Organization guidelines for transmission risk control (World Health Organization, 2009) and the methods are verified for COVID-19 transmission modeling in various studies (Foster and Kinzel, 2021; Sun and Zhai, 2020).

It is worth mentioning that q , the quanta generation rate (quanta/hour), is not a physical unit. Instead, it captures the average number of infectious particles that are exhaled to the environment and the infectiousness of the particles. The value of q can be back-calculated from epidemiological studies of outbreak cases using equation (1) (Rudnick and Milton, 2003; Stephens, 2012).

The original Wells-Riley Equation includes Q , the room ventilation

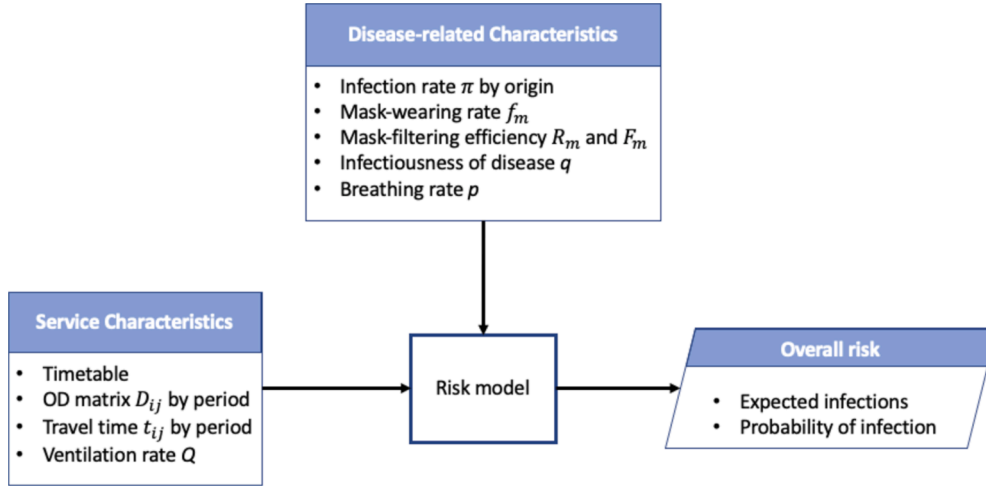


Fig. 1. Transmission risk assessment approach.

rate of clean air (m^3/hour). However, in reality, the aerosols may stay in the environment and decay with time. In other words, there are other mechanisms that can contribute to the concentration and removal of pathogenic material from the air, including the viral inactivation (biological decay in infectiousness of the pathogen) and viral deposition on the ground or on other surfaces (Edwards et al., 2024). This can be incorporated into the model by replacing the Q in equation (1) with a more general removal rate R (m^3/hour) (Miller et al. (2021) and Edwards et al. (2024)), and we have

$$R = Q + vr_i + vr_d \quad (2)$$

where v is the volume of the space (m^3), r_i is the viral inactivation rate ($1/\text{hour}$), r_d is the viral deposition rate ($1/\text{hour}$).

And we can have

$$P = 1 - \exp\left(-\frac{I_p q t}{R}\right) \quad (3)$$

For an empty train arrived at the designed terminal and ready for its next trip, the remaining and decaying virus load and the associated transmission risk in the train car can be considered using equation (3).

However, due to the difficulties to obtain and calibrate the value of r_i and r_d , in this study, we only consider Q the ventilation rate of clean air (m^3/hour), thus the equation (1) will be adopted.

To incorporate additional factors, including mask-wearing and mask filtration efficiency, the original model (equation (1)) has been modified and further developed in the literature (Fennelly and Nardell, 1998; Furuya, 2007; Zhou and Koutsopoulos, 2021). Considering the environment of a train cabin, with both a fraction of infectors and susceptible persons wearing a mask, the probability of infection can be expressed as:

$$P = 1 - \exp\left(-\frac{I(F_m \times p)(R_m \times q)t}{Q}\right) \quad (4)$$

The infectious particles exhaled by infectors are filtered at a certain rate, R_m , the particle penetration rate of the mask. $R_m = 1$ represents the case of no mask wearing where all infectious particles from the infector are exhaled into the air; $R_m = 0$ means a mask with 100 % particle blockage. Similarly, the particles inhaled by susceptible persons could be filtered at a certain rate F_m ($F_m = 1$ means no mask, and $F_m = 0$ 100 % particle blockage).

Modeling infection risk for transit operations

Let us consider a passenger who travels from station i to station j .

During their trip they interact with passengers traveling from r to s if $r < j$ and $s > i$. Conditional on the number of infectors n_{rs} from station r to station s , the probability that a passenger from i to j is not infected by the n_{rs} infected passengers who travel from r to s is given by:

$$P_{ij}^{rs}(\text{notinfected}|n_{rs}) = \exp\left(-\frac{n_{rs} q p t_{ij}^{rs}}{Q}\right) \quad (5)$$

where t_{ij}^{rs} is the time a passenger from i to j is exposed to passengers from r to s (based on the overlap between the two OD pairs).

Infection rate is the percentage of the population that is infected. The number of infectors on OD pair rs , assuming that the probability of a person being infected is π (infection rate), follows the binomial distribution with parameters N_{rs} (the total number of passengers traveling from r to s) and π . Thus, the probability that a passenger traveling on OD pair ij will not be infected by passengers from station r to s is given by:

$$P_{ij}^{rs}(\text{notinfected}) = \sum_{n_{rs}=0}^{N_{rs}} \exp\left(-\frac{n_{rs} q p t_{ij}^{rs}}{Q}\right) \binom{N_{rs}}{n_{rs}} \pi^{n_{rs}} (1 - \pi)^{N_{rs} - n_{rs}} \quad (6)$$

n_{rs} is the number of infectors (carriers). N_{rs} is the number of passengers aboard a train from station r to station s , π the infection rate in the population. π can have different values for passenger groups depending on their origin/destination, in order to accurately capture the spatial pattern of the spread of the virus.

For large values of N_{rs} and small π , the binomial distribution can be approximated using the Poisson distribution:

$$P(n_{rs}) = \binom{N_{rs}}{n_{rs}} \pi^{n_{rs}} (1 - \pi)^{N_{rs} - n_{rs}} \approx \exp(-N_{rs} \pi) \frac{(N_{rs} \pi)^{n_{rs}}}{n_{rs}!} \quad (7)$$

Hence, the probability that a passenger traveling from i to j will not be infected by passengers traveling from r to s is given by:

$$P_{ij}^{rs}(\text{notinfected}) = \sum_{n_{rs}=0}^{N_{rs}} \exp\left(-\frac{n_{rs} q p t_{ij}^{rs}}{Q}\right) \exp(-N_{rs} \pi) \frac{(N_{rs} \pi)^{n_{rs}}}{n_{rs}!} \quad (8)$$

Assuming independence from station to station, the probability P_{ij} of a passenger traveling on the OD pair ij being infected is:

$$P_{ij} = 1 - \prod_{rs} \left[\sum_{n_{rs}=0}^{N_{rs}} \exp\left(-\frac{n_{rs} q p t_{ij}^{rs}}{Q}\right) \exp(-N_{rs} \pi) \frac{(N_{rs} \pi)^{n_{rs}}}{n_{rs}!} \right] \quad (9)$$

Following Zhou and Koutsopoulos (2021), the model can be extended to accommodate mask-wearing behavior. Let f_m be the fraction of mask-wearing passengers. The probability $\bar{P}_{ij}^m$ of a susceptible person

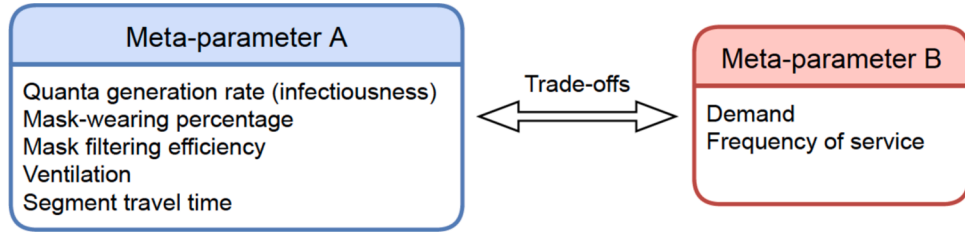


Fig. 2. Meta-parameters A and B.

traveling from i to j and wearing a mask not infected by (masked and non-masked) infectors is calculated by combining equations (7), (8) and (9):

$$\begin{aligned} \bar{P}_{ij}^m &= \prod_{rs} \left[\sum_{n_{rs}=0}^{N_{rs}} \exp\left(-\frac{n_{rs} f_m (F_m \times p) (R_m \times q) t_{ij}^{rs}}{Q}\right) \exp(-N_{rs} \pi) \frac{(N_{rs} \pi)^{n_{rs}}}{n_{rs}!} \right] \times \prod_{rs} \left[\sum_{n_{rs}=0}^{N_{rs}} \exp\left(-\frac{n_{rs} (1-f_m) (F_m \times p) (q) t_{ij}^{rs}}{Q}\right) \exp(-N_{rs} \pi) \frac{(N_{rs} \pi)^{n_{rs}}}{n_{rs}!} \right] \\ &= \prod_{rs} \left[\sum_{n_{rs}=0}^{N_{rs}} \exp\left(-\frac{n_{rs} f_m (F_m \times p) (R_m \times q) t_{ij}^{rs} + n_{rs} (1-f_m) (F_m \times p) (q) t_{ij}^{rs}}{Q}\right) \exp(-N_{rs} \pi) \frac{(N_{rs} \pi)^{n_{rs}}}{n_{rs}!} \right] \end{aligned} \quad (10)$$

$n_{rs} f_m$ is the masked proportion of n_{rs} . R_m , as stated earlier, is the effectiveness of a mask.

Similarly,

can provide interesting insights for policymakers with regard to what can be implemented to lower transmission risk in public transportation systems. Similarly, the importance of virus related characteristics and

their interactions with service characteristics are of interest. These factors can be categorized into two broad categories: viral load and passenger load factors. Factors directly impacting the viral load are mask-

$$\bar{P}_{ij}^{n-m} = \prod_{rs} \left[\sum_{n_{rs}=0}^{N_{rs}} \exp\left(-\frac{n_{rs} f_m (1 \times p) (R_m \times q) t_{ij}^{rs} + n_{rs} (1-f_m) (1 \times p) (q) t_{ij}^{rs}}{Q}\right) \exp(-N_{rs} \pi) \frac{(N_{rs} \pi)^{n_{rs}}}{n_{rs}!} \right] \quad (11)$$

where $\bar{P}_{ij}^{n-m}$ is the probability that a non-masked susceptible passenger traveling from station i to station j is not infected.

Consequently, the probability of infection for a passenger traveling from i to j is given by:

$$P_{ij}^m = 1 - \bar{P}_{ij}^m \quad (12)$$

$$P_{ij}^{n-m} = 1 - \bar{P}_{ij}^{n-m} \quad (13)$$

Using equations (10), (11), (12), and (13), r_{ij} , the expected number of infected passengers traveling on OD pair ij per trip is given by:

$$r_{ij} = P_{ij}^{n-m} (1 - f_m) D_{ij} + P_{ij}^m f_m D_{ij} \quad (14)$$

D_{ij} is the OD flow from i to j (susceptible).

Based on the above equations, the expected number of infections at the system level, r , and the overall probability of infection can be calculated taking into account (weighted by) the distribution of the trips in the system. These metrics are useful measures of risk and can be used to assess the impact of operating characteristics, as well as the effectiveness of different mitigation strategies related to mask wearing, ventilation, etc., on a consistent basis.

Fig. 1 summarizes the risk calculation approach as a function of service characteristics and virus related parameters. The model receives as input the timetable, OD demand matrix, and train travel times between stations. Given the fraction of infectors in the general population (could vary by station) and the fraction of mask-wearing passengers, the model calculates the infection risk.

The impact from factors such as mask-wearing, ventilation rates, travel times, OD demand levels, especially the trade-offs between them,

wearing, mask filtration efficiency, ventilation, and OD travel times. Demand (OD flows) and frequency of service are passenger load related factors.

In the discussion that follows, the superscript o is used to indicate base conditions in terms of service frequency, demand (OD flows), ventilation, infectiousness, travel time, and mask filtration efficiency. Let q^o be the base quanta generation rate (infectiousness of disease), h^o the base case headway, D^o the base demand rate (passengers per unit time) in terms of OD flows, N_{rs}^o the base case average number of passengers traveling from r to s per trip, Q^o the base case ventilation rate, and $t_{ij}^{rs,o}$ the base case travel time a passenger from i to j is exposed to passengers traveling from r to s . Factors α , β , δ , and ε are used to adjust the base values.

$$q = \alpha q^o \quad (15)$$

$$h = \beta h^o \quad (16)$$

$$D = \gamma D^o \quad (17)$$

$$t_{ij}^{rs} = \delta t_{ij}^{rs,o} \quad (18)$$

$$Q = \varepsilon Q^o \quad (19)$$

Consequently, N_{rs} , the average number of passengers per trip from r to s , as a function of the corresponding average base number of passengers N_{rs}^o per trip, is given by:

$$N_{rs} = \beta \gamma N_{rs}^o \quad (20)$$

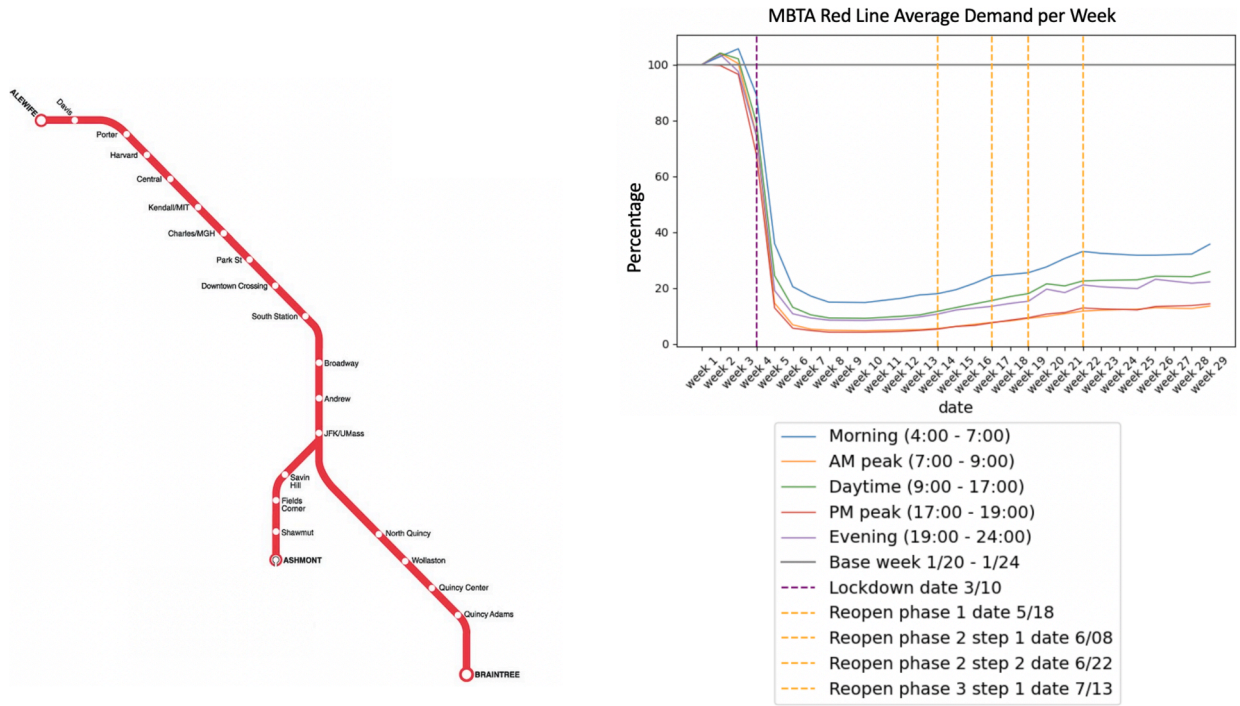


Fig. 3. MBTA Red Line average demand per week after 2020/02/16. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The probability of infection (equation (12)) can then be expressed as a function of the base case conditions and the scaling factors.

$$\begin{aligned}
 p_{i,j}^m &= 1 - \prod_{rs} \left[\sum_{n_{rs}=0}^{BN_{rs}^0} \exp \left(- \frac{n_{rs}(F_m \times p)(1 - f_m(1 - R_m))(\alpha \times q^0) \delta t_{ij}^{rs0}}{\varepsilon Q^0} \right) \exp \left(- \beta \gamma N_{rs}^0 \pi \right) \frac{(\beta \gamma N_{rs}^0 \pi)^{n_{rs}}}{n_{rs}!} \right] \\
 &= 1 - \prod_{rs} \left[\sum_{n_{rs}=0}^{BN_{rs}^0} \exp \left(- A \frac{n_{rs}(p)(q^0) t_{ij}^{rs0}}{Q^0} - BN_{rs}^0 \pi \right) \frac{(BN_{rs}^0 \pi)^{n_{rs}}}{n_{rs}!} \right] \quad (21)
 \end{aligned}$$

where,

Table 1

Base case parameters and assumptions.

- 1 Mask effectiveness in terms of particle penetration rate for exhaling (inhaling) $R_m = 0.5$ ($F_m = 0.5$), as used and reported in (Pan et al., 2020; Stutt et al., 2020), to model normal quality face coverings.
- 2 Mask-wearing percentage f_m in the population is assumed to be 0 %.
- 3 A q value = 270 quanta/hour is assumed, which is also consistent with the findings from studies, as over 100 quanta/hour (Buonanno et al., 2020a; Buonanno et al., 2020b; Miller et al., 2021). The value is back-calculated using the method mentioned in studies (Rudnick and Milton, 2003; Stephens, 2012) by applying equation (1). Two COVID-19 outbreak cases in public transportation vehicles reported in the literature (Luo et al., 2020) are used to derive the q value.
- 4 Ventilation rate in a cabin of a subway train $Q = 1958 \text{ m}^3/\text{hour}$. The ventilation rate is derived based on Massachusetts Bay Transportation Authority MBTA Red and Orange Line New Vehicle Procurement Technical Provisions (Massachusetts Bay Transportation Authority, 2014), which states the ventilation system requirement for the new vehicles: 7.5 CFM (0.212 m^3 per minute) per passenger at AW2 train loading (around 154 passenger).
- 5 Breathing rate per person (m^3/hour), $p = 0.72 \text{ m}^3/\text{hour}$ (Environmental Protection Agency, 2015).
- 6 According to the published schedules, the scheduled headways from the branches are 9 min, coordinated to 4.5 min headways in the trunk.
- 7 Average infection rates in the population are assumed to be 0.92 % for all stations according to data as of June 2021 in the Greater Boston area.
- 8 The base case demand is assumed to be equal to September 2020 demand,

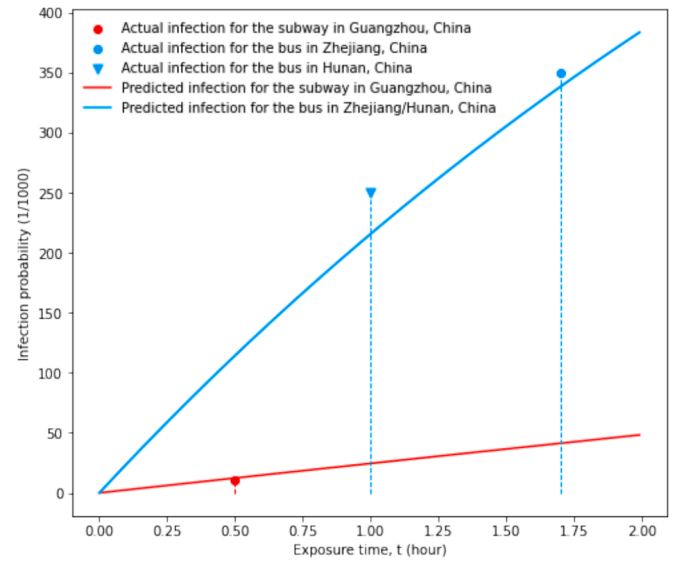


Fig. 4. The predicted and actual probability of infection in existing cases.

$$A = \frac{\alpha \delta}{\varepsilon} (1 - f_m(1 - R_m)) F_m \quad (22)$$

$$B = \beta \gamma \quad (23)$$

Meta-parameter A represents the combined effect of the factors that affect the airborne viral load, and B the factors that affect the passenger (carrier) load, as shown in Fig. 2. They can be used to calculate the change in passenger load required to compensate for an increase in viral load, for example, due to an increase in the infectiousness of the disease. It should also be noted that A is impacted mainly by external factors (e.g. mask-wearing), while B is a function of demand characteristics and operating policy (e.g. frequency of service).

Table 2
Comparison to various studies.

Case	Exposure time (hour)	Ventilation rate (m ³ /hour)	f_m	Number of infectors (carrier)	Number of passengers onboard	Actual probability of infection	Predicted probability of infection
Subway in Guangzhou, China (China Daily, 2022)	0.5	2000 (according to China's standards for metro vehicles GB/T 7928–2003)	100 % (China's mask mandate)	1	228 (estimated according to China's standards for metro construction 104–2008)	0.01	0.01
Bus in Zhenjiang, China (Shen et al., 2020)	1.7	800 (according to China's standard for bus vehicles CJT 134–2001)	0 % (No mask mandate)	1	68	0.35	0.34
Bus in Hunan, China (Sun and Zhai, 2020)	1	800 (according to China's standard for bus vehicles CJT 134–2001)	0 % (No mask mandate)	1	12	0.25	0.22

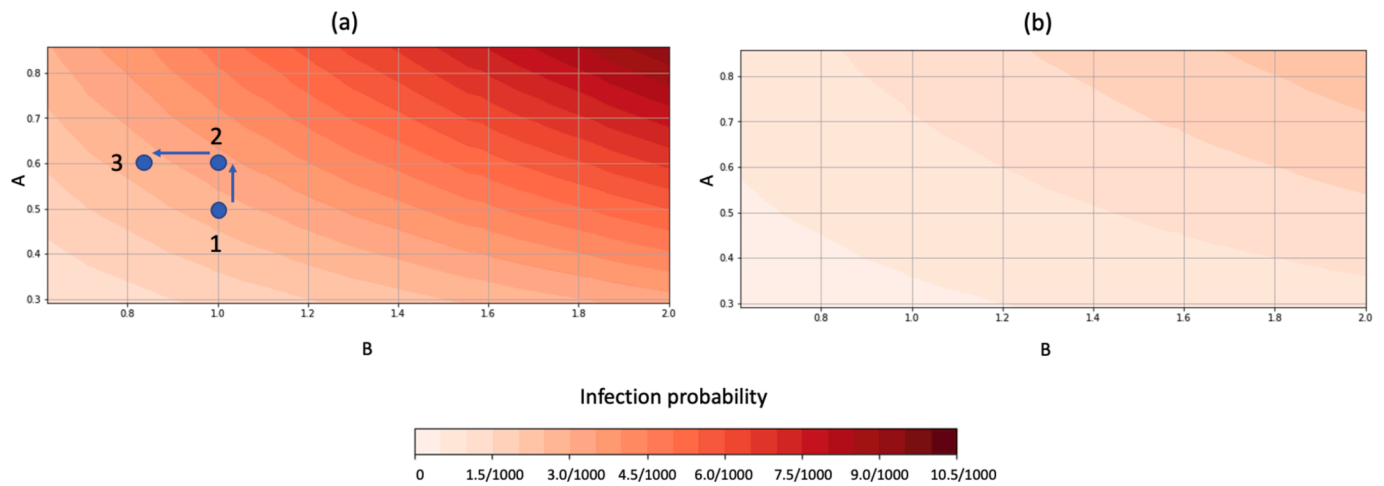


Fig. 5. Infection probability iso-curves under various meta-parameter A and B values. (a) Infection probability for unmasked passengers; (b) Infection probability for masked passengers.

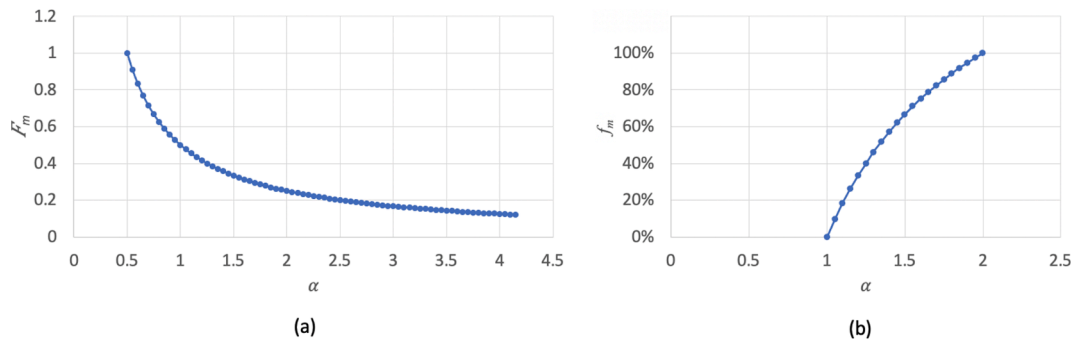


Fig. 6. Interactions between factors from meta-parameter A: (a) α vs. F_m , (b) α vs. f_m , ($A = 0.5$).

Trade-offs and interactions among factors related to meta-parameters A and B, can also be explored. Agencies may be interested in maintaining a target risk level. As factors change, for example infectiousness of a virus due to new variants, equations (22) and (23) can be used to assess quickly the changes in operating strategies that may help mitigate the increase in risk. For example, $\alpha = 1.5$ (infectiousness increased by 50 %) implies that $A = 1.5 A^0$ (the base value). Required changes in F_m to reduce risk to the original level can then be assessed. Similarly, changes in headways (value of β) can also be assessed.

Application

The MBTA Red Line (Fig. 3) provides the background for the case

study. The Red Line is the highest frequency and ridership heavy rail transit line in the MBTA network with an average weekday ridership of 251,000 passengers (Massachusetts Bay Transportation Authority, 2019). The line is composed of a trunk section and two branches on the south end. The trunk section connects the north terminus (Alewife Station) to JFK/UMass station and is 14.2 km (8.8 miles) long with 13 stations. The Braintree branch is 14.2 km (8.8 miles) long with 5 stations and the Ashmont branch 4.7 km (2.9 miles) long with 4 stations.

The demand on the Red Line was greatly impacted by the COVID-19 pandemic as Fig. 3 indicates. The analysis focuses on the evening peak period (16:00 to 18:30) using September 2020 demand levels (OD flows in 15-minute intervals) and the typical schedule. The direction we consider is the Southbound direction.

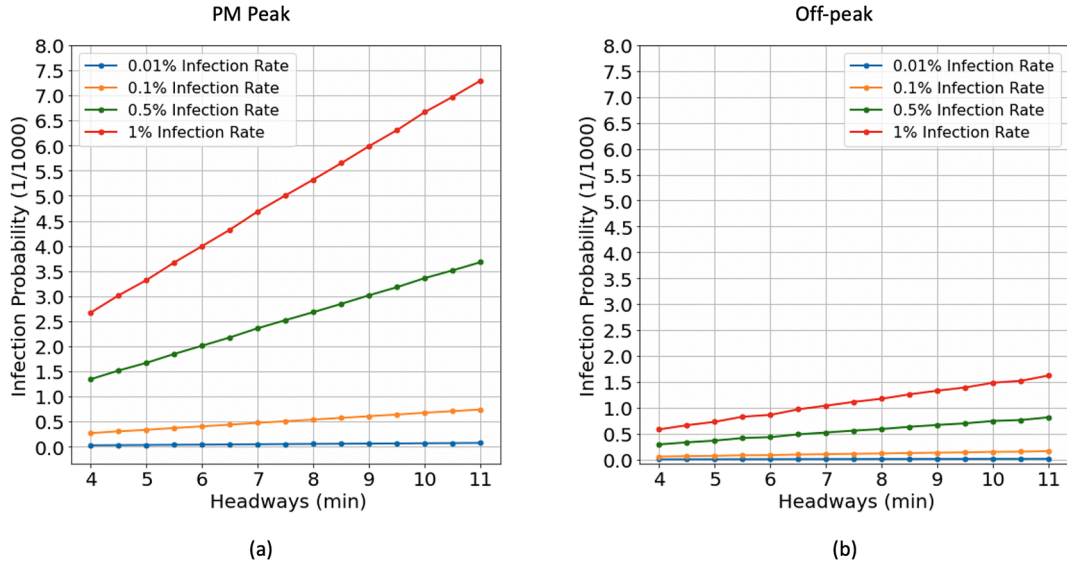


Fig. 7. Infection probability under different headways and infection rates (a) PM peak. (b) Off-peak.

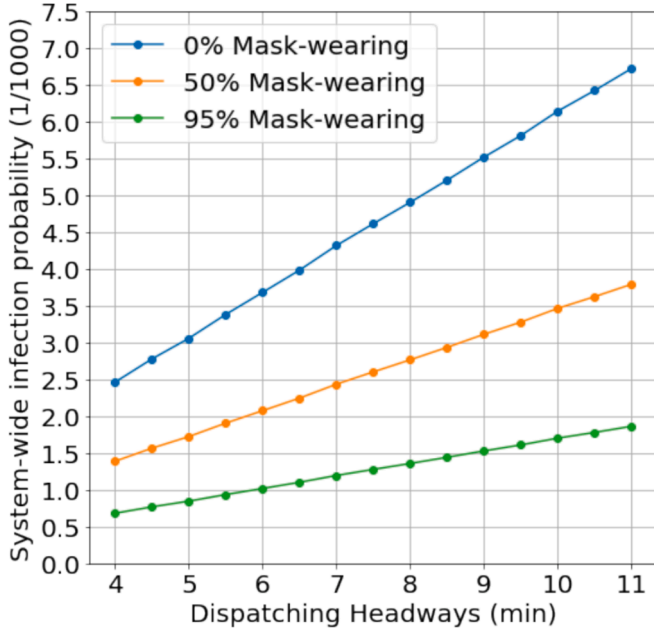


Fig. 8. Infection probability under different headways and mask-wearing proportions during the PM peak.

Experimental design and assumptions

Table 1 summarizes the base parameter values for the experiments.

Regarding the ventilation rate of train cars, the current assumption for train ventilation is based on the technical specifications of one model of train cars from MBTA, which does not represent the entire operating fleet. In future studies, we propose incorporating more precise ventilation data for individual train trips, as the modified Wells-Riley model we employed is based on each train trip, and each train trip is associated with a specific train type. Non-air-conditioned vehicles, for instance, may provide enhanced ventilation when windows are open. To improve the accuracy of the model, the ventilation rate of clean air (Q , m³/hour) in equation (1) could be updated for each train trip based on the specific train type's ventilation system and its performance. This can be done either through direct measurement of the ventilation rate or by

consulting relevant literature for each vehicle type.

Based on the above assumptions, the base case system-wide infection probability is 2.8/1000. The base case assumes that passengers do not wear a mask ($f_m = 0\%$). For the calculation, we also assume that the air exchange by door-opening at stations is neglected and risk due to waiting at stations is not considered.

The objective of the case study is to apply the proposed model to assess transmission risk in subway systems and, explore the trade-offs between viral load and passenger load *meta*-parameters. The impact on transmission risk of headways, spatial patterns of infection rates (as opposed to assuming uniform rates across the service area), and passenger distribution on train cars, is also of interest.

Results and discussion

Model verification

This study has collected data from several actual pandemic cases for validation and verification of the proposed modified Wells-Riley equation, especially its accuracy in applications in the transportation environment. Using similar methods as Sun and Zhai (2020), we compared the actual percentage of infection from several actual infection cases in subway and buses (China Daily, 2022; Shen et al., 2020; Sun and Zhai, 2020) with the model's predicted infection probability in Fig. 4. The breathing rate p , mask effectiveness F_m and R_m , and q values are the same as in Table 1. The parameters of actual pandemic cases can be seen in Table 2.

Table 2 and Fig. 4 show the predicted infection probability from the modified Wells-Riley model, compared against the actual infection rates from three public transportation related infection cases (China Daily, 2022; Shen et al., 2020; Sun and Zhai, 2020). The comparison shows that the modified model has reasonable prediction accuracy.

Meta-parameters A and B trade-offs

The latest variants of COVID-19 are more infectious than the original strain, for example, the Delta variant is around two to five times as infectious as previous variants (Centers for Disease Control and Prevention, 2021; Herrero, 2021). When the infectiousness of the disease increases, in principle, the impact can be mitigated by changing other factors that contribute to the value of A , or by lowering the value of B (representing passenger load) through changes in the frequency of service.

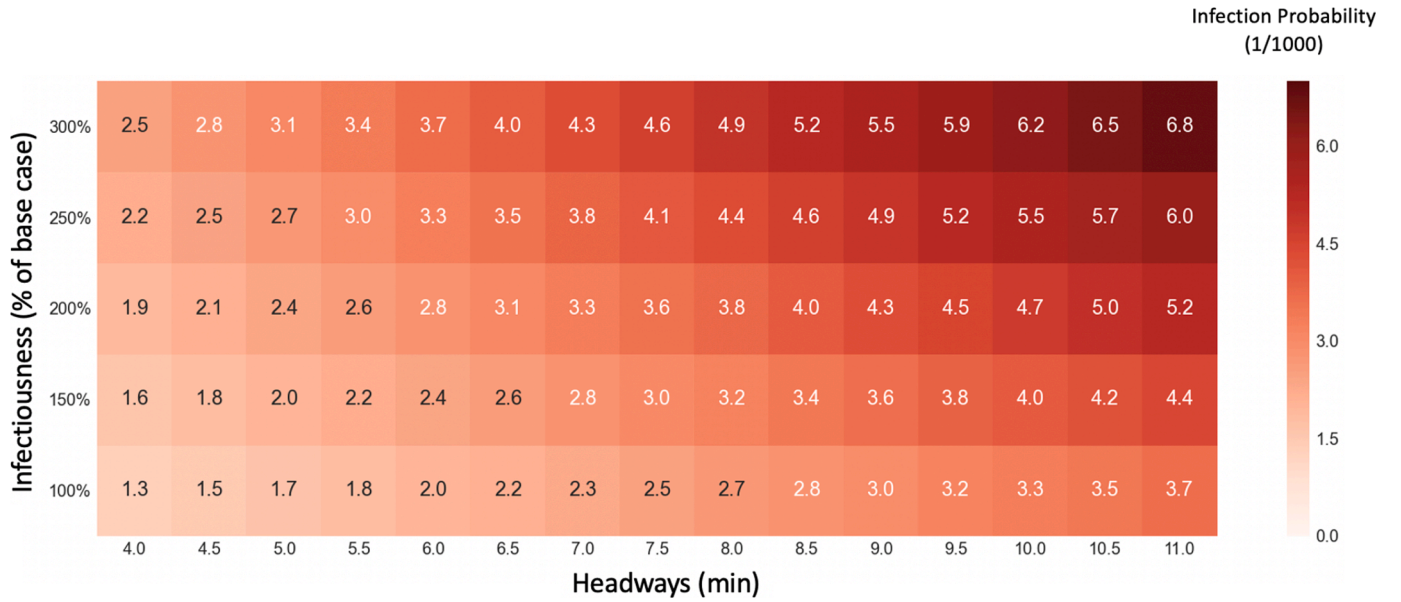


Fig. 9. System-wide infection probabilities (in 1/1000) under various headways and infectiousness.

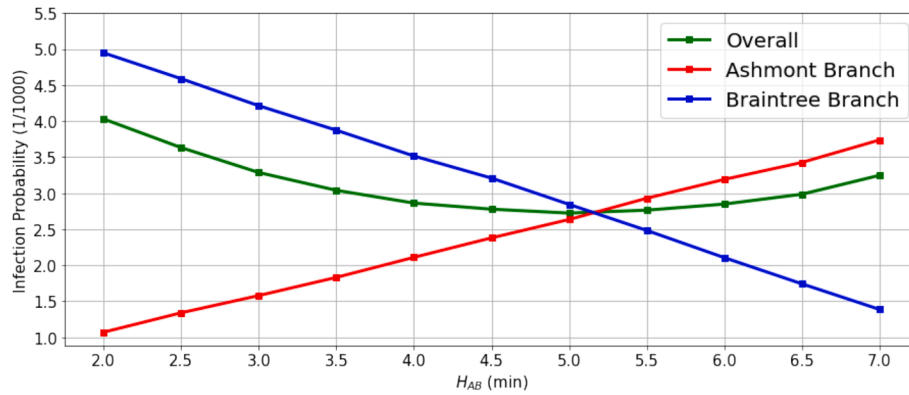


Fig. 10. Infection probability under various headways between Ashmont and Braintree trains (H_{AB}).

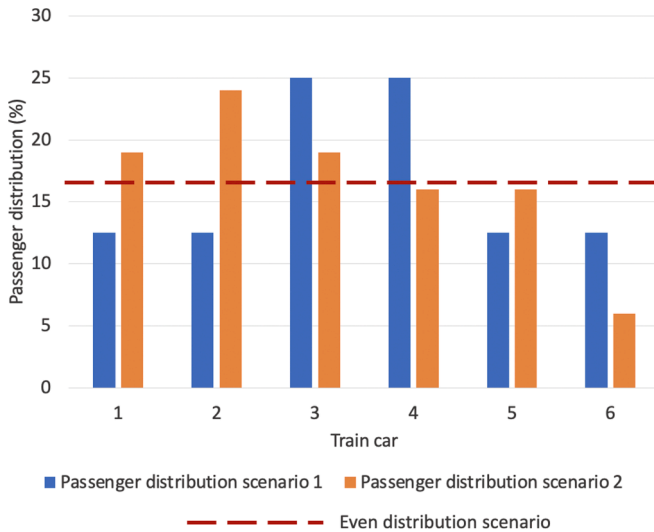


Fig. 11. Passenger distribution among cars, scenarios 1 and 2, and even distribution.

Fig. 5a shows a heatmap of the system-wide probability of infection for unmasked susceptible passengers, for various values of the viral load *meta*-parameter *A* and passenger load *meta*-parameter *B*. The lines represent the *iso*-risk curves for different values of *meta*-parameters *A* and *B*. As expected, the system level overall infection probability increases as the values of *A* and *B* increase. However, *meta*-parameters *A* and *B* have different effects on transmission risk. Fig. 5a can be a useful tool to quickly assess desired targets to mitigate impacts on risk due to changes in external factors. Let us consider the base case (overall infection probability = 2.8/1000) where *A* = 0.5 and *B* = 1.0 (point 1 in Fig. 5a). A 20 % increase in the virus infectiousness (for example due to mutation) means that *A* = 0.6 (point 2 in Fig. 5a). The change in the *B* parameter to mitigate the risk and bring it to previous levels requires passenger load changes that result in a new *B* value of around 0.85 (point 3). Since *B* is a function of frequency of service and OD flow patterns, and assuming that demand remains the same, headways have to change by a factor $\beta = 0.85$ (reduced) so that the risk returns to its original value. However, depending on the base operations, this may not be easily accomplished, given line (signal) capacity constraints. Alternatively, operators and policy makers, may attempt to mitigate the increased risk, due to the increased infectiousness, by changing other parameters that contribute directly to *A*. α is the factor that adjusts the scale for the base quanta generation rate q^0 ($q = \alpha q^0$), i.e. the infectiousness of the disease (it contributes to the value of *A*). To fully

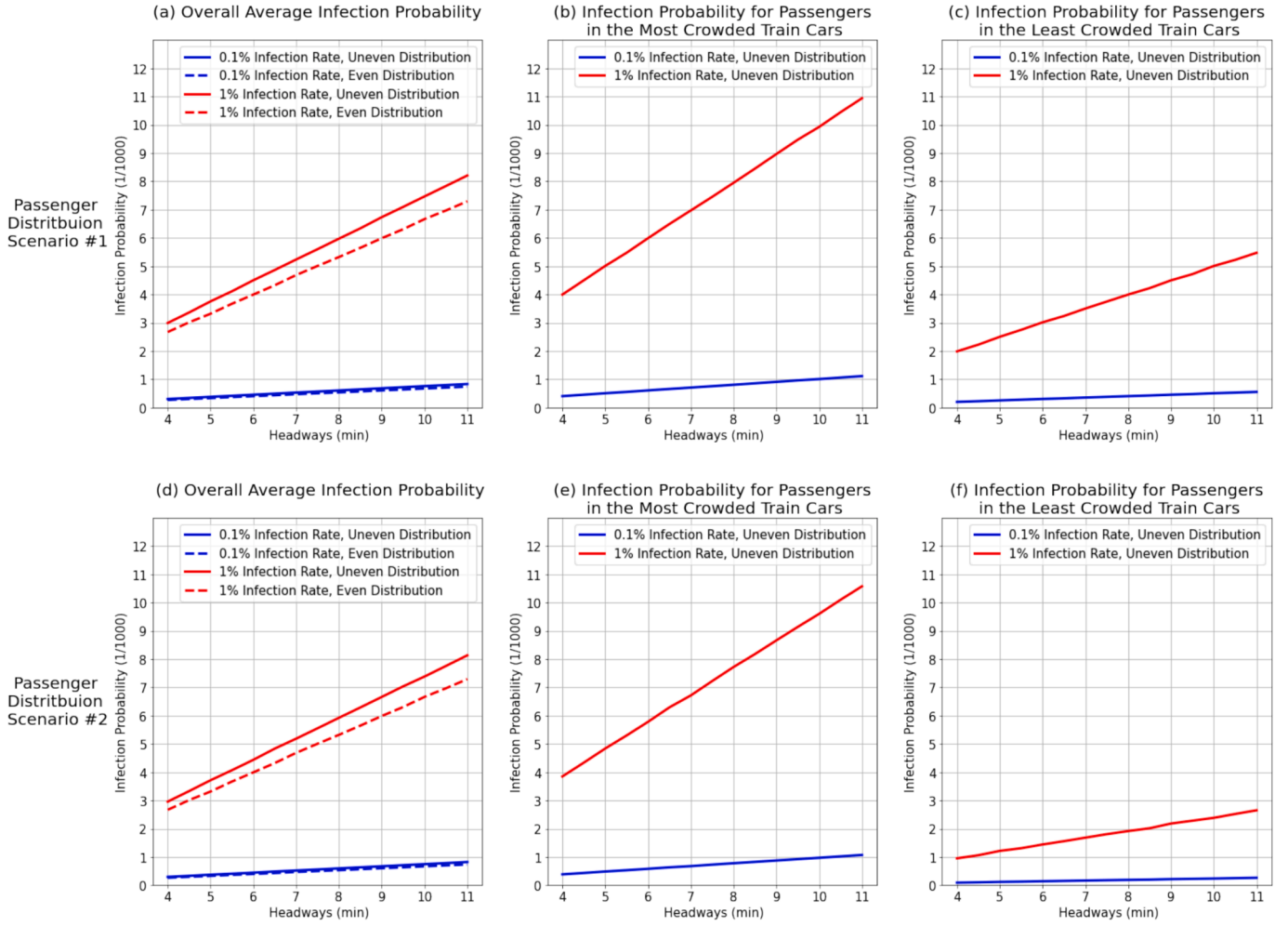


Fig. 12. Infection probability for different passenger distribution scenarios under various dispatching headways and infection rates.

compensate the increase in the infectiousness, other factors contributing to the value of A could also be changed to keep its value at the same level (assuming B remains unchanged). For example, agencies can encourage mask wearing to increase the proportion f_m in the population by distributing masks with better protection, such as N95 or surgical masks (Tirupathi et al., 2020). Fig. 5b shows a heatmap of the system-wide probability of infection for masked susceptible passengers, for various values of the viral load *meta*-parameter A and passenger load *meta*-parameter B . The results indicate that the infection risk is significantly lower for masked passengers compared to unmasked passengers.

In the base case, $A = 0.5$. A is a function of various viral load parameters ($A = \frac{\alpha\beta}{\epsilon}(1 - f_m(1 - R_m))F_m$). Fig. 6 illustrates the trade-offs among the factors contributing to *meta*-parameter A so its value remains at $A = 0.5$. In particular, the figure illustrates the combination of values of α (infectiousness) and F_m (mask effectiveness) and α and f_m (mask wearing percentage) so that $A = 0.5$. Fig. 6a shows the interactions between infectiousness (α) and mask effectiveness (F_m). The results indicate that F_m should decrease (mask efficiency should increase) as the value α increases. For example, a 2 times increase in infectiousness requires masks that can filter out approximately 75 % of the particles (from the base case of 50 %). Fig. 6b shows the required increase in the proportion of mask-wearing passengers to compensate for an increase in infectiousness (α) (so that the value of A remains the same at 0.5). A two times increase in infectiousness requires that mask wearing percentage increases to 100 % (assuming mask-wearing effectiveness remains at 50 %). However, increasing the mask-wearing proportion can compensate at most 2 times increase in infectiousness. If

the infectiousness increases by more than 2 times, mask wearing proportion alone cannot bring the overall risk back to the base level (even if all passengers wear a mask).

Impact of operating characteristics

It has been shown that operating strategies (headways) and service reliability impact the transmission risk (Zhou and Koutsopoulos, 2021). Headways impact the trainload (and passenger load related *meta*-parameter B) and subsequently the transmission risk. Fig. 7 summarizes the impact on the system level risk of various infection rates at different scheduled headways (x-axis). For other parameters not stated in the legends, the values are as stated in Table 1 (base case).

Fig. 7a shows the impact on the infection probability during the PM peak under various headways. When the population infection rate is lower than 0.1 %, the risk is relatively low, and the differences among the various headway levels are small. As the infection rate increases from 0.1 % to 1.0 %, the risk increases significantly. Fig. 7b shows the impact on the infection probability of various headways during off-peak hours. The average off-peak hourly demand is 32 % of the peak hourly demand. Given the lower demand levels in the off-peak, the risk is much lower and relatively less sensitive to increases in infection rates compared to the PM peak case.

Fig. 8 summarizes the impact on risk at the system level of mask-wearing at different headways (x-axis) during the PM peak. For other parameters not stated in the legends explicitly, the values are as stated in the base case.

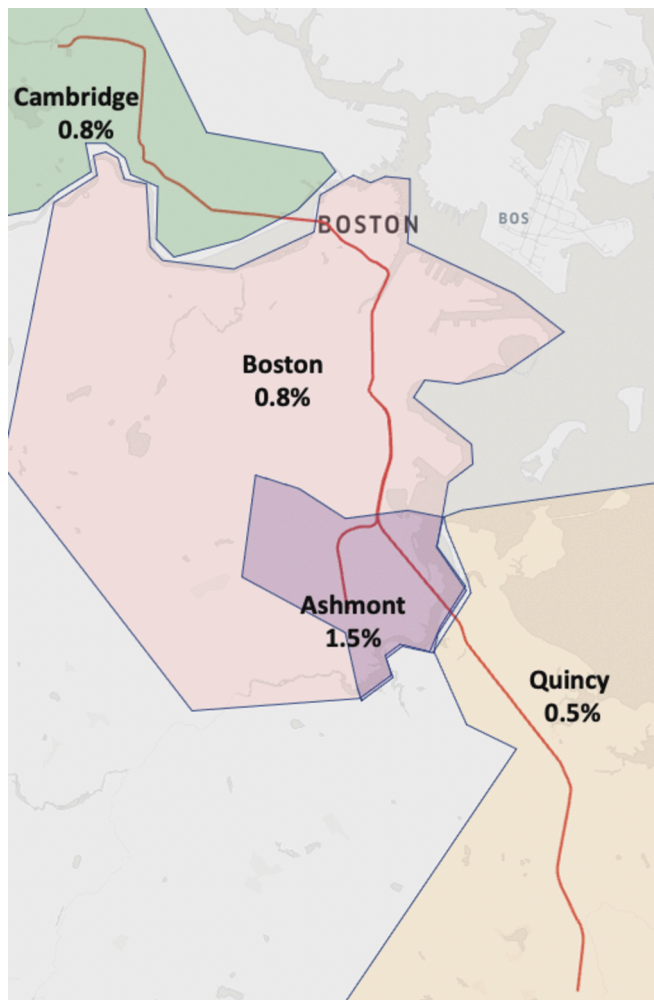


Fig. 13. Infection rates in different municipalities and neighborhoods served by the Red Line. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

As shown in Fig. 8, when passengers wear masks, the infection probability is not as sensitive to the service headways compared to the non-masked case. With 50 % of passengers wearing masks, the infection probability decreases by approximately 44 % from the base case. With 95 % of passengers wearing masks, the infection probability decreases by approximately 73 % from the base case. The mitigation effect of mask-wearing is significant.

Fig. 9 shows a heatmap of the system-wide infection probability P (expressed in units of 1 over 1000) under various combinations of headways and level of infectiousness. The results indicate that increasing the frequency of service can mitigate transmission risk and compensate for a rise in infectiousness of the disease. For example, when the headway is 9 min and the infectiousness is at the base case value, the infection probability is 5.52/1000. When the virus is 2 times more infectious than the base case, agencies need to adjust the headways to lower than 6.0 min to compensate for such a rise in infectiousness.

When designing headways during a pandemic, it is also important to take the spatial demand patterns into account. In the MBTA Red Line case, the passenger demand travelling to different branches varies. Braintree (B) trains are carrying more passengers to their branch than Ashmont (A) trains. The Braintree branch is also longer in distance, so its passengers have longer exposure times. During normal operations, trains to Braintree and Ashmont dispatched at the trunk terminal (Alewife) alternate (A-B-A-B), and the headways between consecutive trains are equal. Since Ashmont demand is lower, equal headways between

consecutive trains result in Ashmont trains having smaller loads compared to Braintree trains.

Using the transmission modeling framework developed in the previous section, we explore the impact of headway allocation between the branches on transmission risk. The settings are consistent with the base-case assumptions, except the headway allocation between branches. Let H_{AB} be the headway of an Ashmont branch train with the Braintree train in front of it (H_{BA} is the headway of a Braintree branch train with the Ashmont train in front of it). The headways in the branches are assumed to be the same as in the base case, 9 min. Under current practice, the headways in the trunk section are 4.5 min. Given the unequal demand of the two branches and in order to assess the impact of headway allocation, we assume that H_{AB} ranges from 2.0 to 7.0 min, while H_{BA} assumes a value of $9.0 - H_{AB}$ (which maintains the 9.0 min headways in the branches).

Fig. 10 illustrates the risk at the system and branch levels as a function of H_{AB} . The infection probability for Braintree trains decreases as H_{AB} increases from 2.0 min to 7.0 min. At the same time, the infection probability for Ashmont trains increases. With the uneven allocation of headways, Braintree trains have fewer trunk passengers as their headway from the Ashmont train in front decreases. As Fig. 10 shows, even headway allocation between branches, $H_{AB} = H_{BA} = 4.5$ min, is not the optimal allocation with respect to the overall transmission risk (assuming uniform infection rate in the service area). The overall infection probability reaches its lowest value at $H_{AB} = 5$ min and $H_{BA} = 4$ min. Changing the headway allocation can help mitigate transmission risk, lowering the overall infection probability by 3 %.

Load on trains is impacted not only by OD flows and headways, but also by how passengers are distributed in train cars. When trains arrive at platforms, passengers tend to choose the location on the platform and consequently the boarding car, often based on the locations of the platform entrances and access locations (Jenelius and Cebecauer, 2020; Krstanoski, 2014). Uneven passenger distribution on platforms can cause uneven distribution in train cars. Passengers in crowded train cars experience higher transmission risk than passengers in other less crowded train cars. We analyze the impact on risk of the distribution of passengers among the various cars of a train using the Red Line setting. It is assumed that all stations in the network have similar patterns of passenger distribution. Each Red Line train has six cars. Based on results reported in Krstanoski (2014) for the passenger distribution at Bloor Station, Toronto, we consider two scenarios of car load distribution as shown in Fig. 11. Under scenario 1, the 1st, 2nd 5th and 6th cars carry 12.5 % of the passengers each and the 3rd and 4th cars carry 25 % each. Under scenario 2, the 1st and 3rd cars carry 19 % of the passengers, the 2nd car 24 %, the 4th and 5th cars 16 %, and the 6th car 6 % (under even distribution each car carries 16.7 % of the passengers).

The results are shown in Fig. 12. It is assumed that headways range from 4.0 to 11.0 mins and there are two levels of infection rate (0.1 % and 1.0 %). Fig. 12a and 12d show that under passenger distribution scenarios 1 and 2, respectively, when the headways increase, overall system-wide infection probability increases significantly. However, the risk is similar under the two car load distributions. Fig. 12b and 12e show the infection probability for the passengers who are on the most crowded car on each train. The infection probability for passengers on the most crowded car is more than 33 % and 28 % higher (scenarios 1 and 2 respectively) than the overall system-wide infection probability. Fig. 12c and 12f show the infection probability for the passengers who are on the least crowded car. Not surprisingly, the least crowded train cars have a lower infection probability.

It is important to point out that when the passenger distribution among train cars is non-uniform, as in the above cases, the overall system-wide infection probability can be up to 12.5 % higher (depending on the conditions) than when trains have uniform passenger distribution among cars. At low infection rates (e.g. 0.1), infection risk is not sensitive to the load distribution.

Considering these findings, public transport agencies should actively

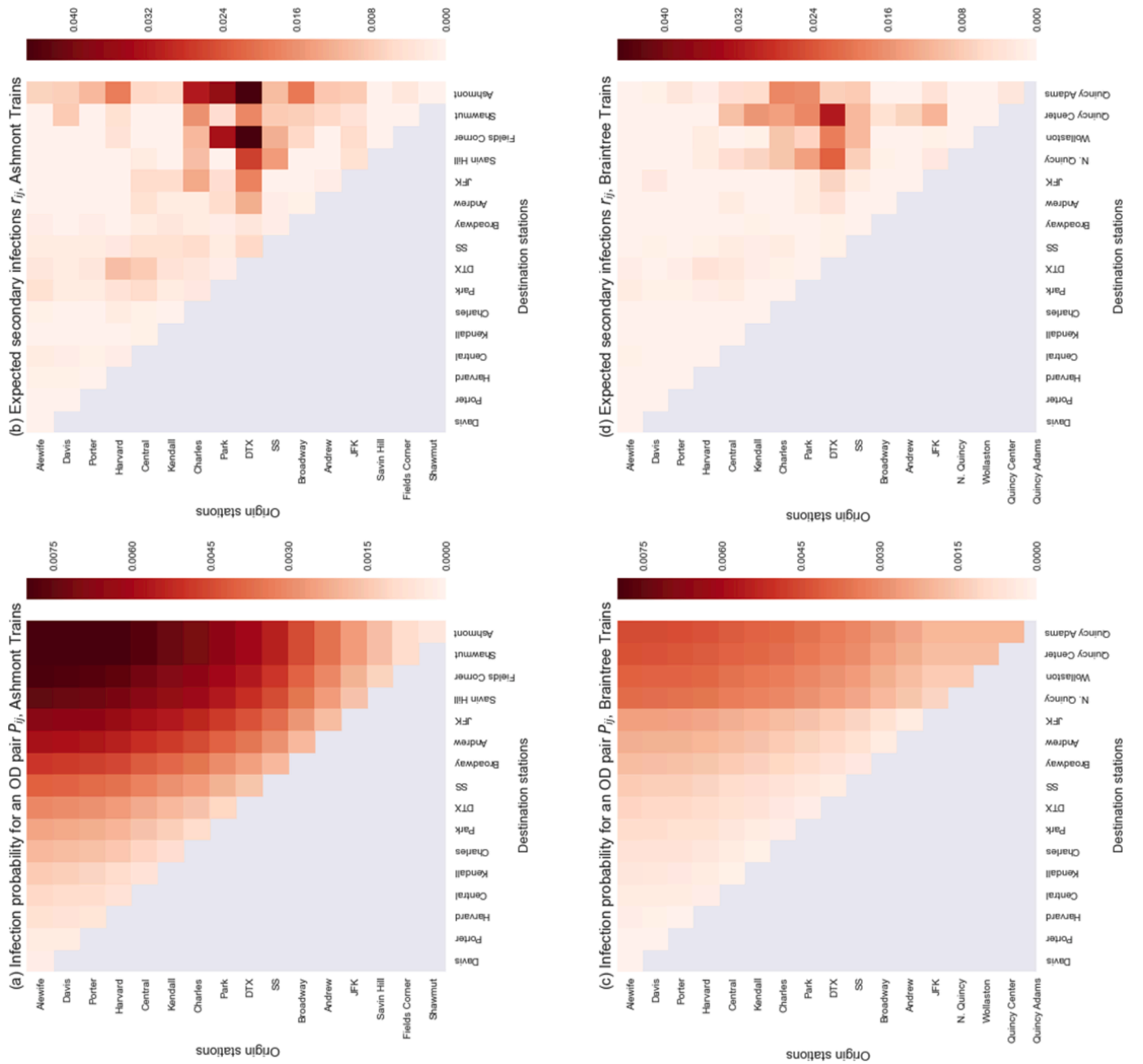


Fig. 14. Infection probability and expected secondary infections (infection rate scenario 1).

manage the passenger distribution on the platforms during a pandemic to maintain relatively even passenger distribution among cars. The results also point out that other policies, where for example, an agency designates a car in the train only for elder and vulnerable users, may be worth examining.

Impact of spatial distribution of infection rates

The previous analyses assumed a uniform infection rate in the population across all stations. However, based on the COVID-19 infection rate data, the infection rates may vary by neighborhood or community, due to sociodemographic characteristics and other reasons (Dooling, 2020). Cambridge's infection rate in January 2021 was 0.8 % (City of Cambridge, 2021), Boston's 0.8 % (City of Boston, 2021), Ashmont's around 1.5 % (Boston Public Health Commission, 2022) and Quincy's 0.5 % (City of Quincy, 2021). The infection rate assumptions are shown in

Fig. 13.

Two scenarios of spatial distribution of infection rates are studied. Scenario 1 assumes that City of Boston and Cambridge's passengers have a uniform 0.8 % infection rate, Braintree branch passengers (PM peak passengers whose destinations are on the Braintree branch) 0.5 %, and Ashmont branch passengers (PM peak passengers whose destinations are on the Ashmont branch) 1.5 % as discussed above. In scenario 2, a uniform 0.92 % infection rate is assumed. The value reflects the average infection rates used in scenario 1 weighted by the passenger demand in each section. The total expected number of carriers is the same as in scenario 1.

Under scenario 1, the spatial pattern of transmission risk for different OD pairs is shown in Fig. 14. In this study, secondary infections are defined as the number of infections caused by the original virus carriers in the train. Fig. 14a and c illustrate infection probabilities by OD pair while Fig. 14b and d show the expected number of infections. Ashmont

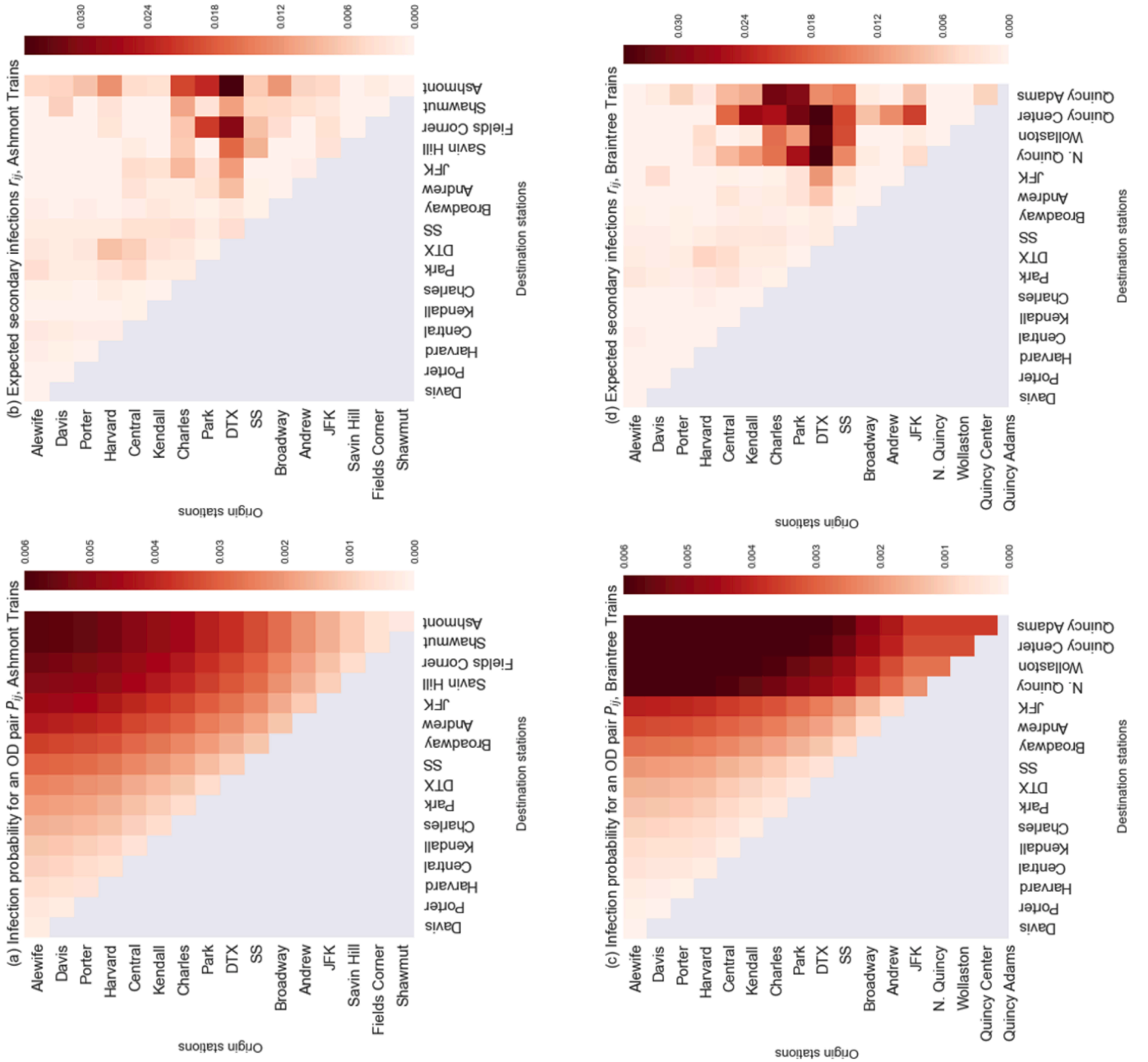


Fig. 15. Infection probability and expected secondary infections (uniform infection rate, scenario 2).

trains have a higher number of infections, even though they have lower demand and smaller train loads than Braintree trains. They also have a higher infection probability than Braintree trains.

Fig. 15 shows results assuming the equivalent uniform infection rate of 0.92 %. In this case, the determining factors for the risk distribution are the demand characteristics and the exposure time. OD pairs on the Braintree branch now have higher infection probabilities and more expected secondary infections, which is expected considering the higher train load and longer travel times (exposure). The system-wide probability of infection (overall transmission risk) under the more realistic spatial distribution of infection rates is 1.5 % higher than under the uniform distribution.

To explore how spatially different infection rates impact the optimal headway allocation between the two branches, we analyze the risk

assuming that the H_{AB} headways range from 2 to 7 mins. As before, $H_{BA} = 9.0 - H_{AB}$. Fig. 16a summarizes the results for the infection rate scenario 1 (actual infection rates). For comparison purposes, the results under infection rate scenario 2 (uniform infection rates) are also shown in Fig. 16b. Based on the results shown in Fig. 16a, the overall infection probability is lowest for $H_{AB} = 3$ min. Considering that the Ashmont branch has lower demand, the optimal allocation reduces the load on Ashmont trains even more, in order to mitigate the impact of higher infection rates in that branch. On the contrary, under scenario 2 with a uniform infection rate, the optimal allocation is $H_{AB} = 5.0$ min which reduces the load on the Braintree trains. The results further highlight the importance of considering the spatial patterns of demand and infection rates when evaluating mitigation strategies.

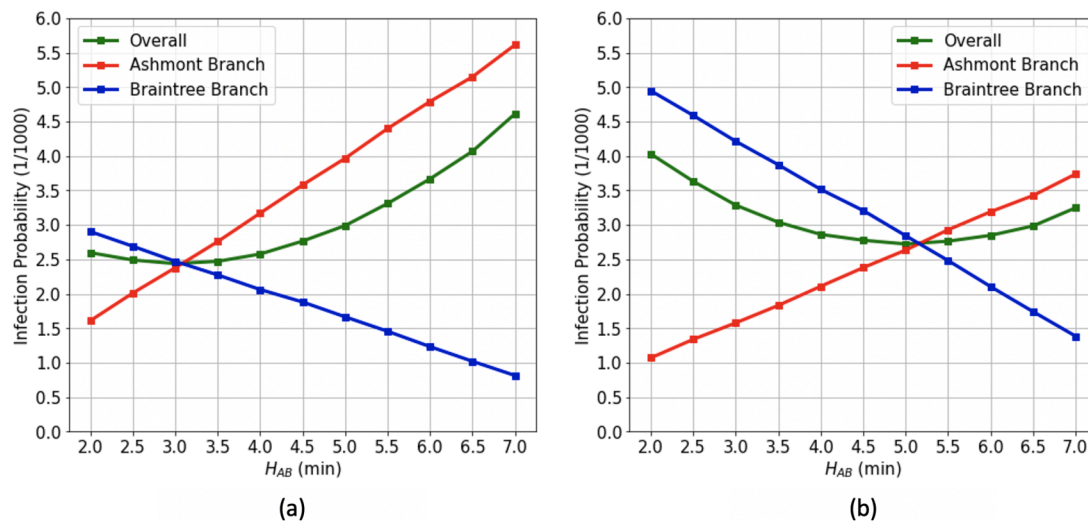


Fig. 16. Infection probability as a function of headway allocation (a) non-uniform spatial distribution of infection rate; (b) equivalent uniform spatial distribution of infection rate.

Discussion and conclusions

In this paper, we propose an airborne transmission modeling framework based on the Wells-Riley model using as inputs transit operating characteristics, OD demand, and virus attributes. The model is sensitive to various factors that an operator can control, external factors related to the characteristics of the virus, as well as factors that may be subject to broader policy decisions (e.g. mask-wearing). It can be used to assess airborne transmission risk as a function of OD flows, actual operations, and factors such as mask-wearing, ventilation, and magnitude and spatial characteristics of infection rates. Using data from the MBTA's Red Line, the study explores the effectiveness of different mitigation methods. Peak period risk increase is more significant with the increase in headways, when compared with the off-peak period. Also, mask-wearing is useful for risk mitigation. Increasing service frequency mitigates airborne transmission risk. Adjusting the headway allocation to balance train loads in lines with branches can reduce the airborne transmission risk. Furthermore, unbalanced passenger distribution on different cars on a train increases the overall system-wide risk. Spatial patterns, especially with respect to infection rates, are important and should be taken into account when designing strategies to mitigate risk.

Several policy implications can be drawn from this case study. From a public health perspective, both transportation and health agencies can encourage mask-wearing as a key measure to reduce transmission risk, particularly in scenarios where infectiousness increases due to the emergence of new variants. Distributing masks with higher protective capacity among passengers can also be a useful mitigation strategy. Furthermore, localized strategies such as deploying staff to encourage mask-wearing in areas with higher infection rates could prove effective. However, it is important to note that as infectiousness increases significantly, mask-wearing alone may not sufficiently reduce transmission risks. A combination of strategies should be implemented, such as increasing train frequency to reduce crowding.

From an operational perspective, train service frequency plays a crucial role in mitigating transmission risk. It is not only the overall frequency of services that matters but also the spatial distribution of demand and infection rates across different neighborhoods and communities. Frequent services can reduce transmission risks, but when passengers are wearing masks, the sensitivity to service frequency is lower than in cases without mask usage. Additionally, headway design during a pandemic should consider spatial demand patterns, especially

on lines with branches that may have varying passenger loads. Adjusting headways to balance train loads can help reduce transmission risks. Another key operational issue highlighted by this study is the uneven passenger distribution among train cars, which increases system-wide infection risk. Balancing the load across all train cars can mitigate this risk. Finally, while increasing frequency may be operationally challenging due to staffing shortages, enforcing mask mandates could be a more feasible approach to significantly lower transmission risks. Equity and accessibility should also be considered when designing service frequencies during pandemics.

This study has several limitations. First, some parameters in the modified Wells-Riley model are based on specific assumptions, and policy evaluations rely on idealized conditions. While the model has been validated with previous cases, further parameter calibration is needed. For example, real-world data on mask-wearing behavior or infection rates in specific neighborhoods may be incomplete, and ventilation assumptions are based on a single train model, which does not represent the entire fleet. Second, the Wells-Riley model itself has limitations, particularly its assumption of a well-mixed air environment. To improve transmission risk assessments, studies can consider combining the Wells-Riley model with Computational Fluid Dynamics (CFD) models for greater accuracy (Wang et al., 2022). Third, as noted in the introduction section, viruses can spread through two primary transmission routes: fomite and aerosol. While COVID-19 has predominantly spread through aerosols, future pandemics may involve other transmission routes. It is important to note that this study focuses exclusively on aerosol transmission, which has been identified as the dominant transmission route for COVID-19. This limitation means that other transmission routes, such as fomite (surface-based) transmission, are not considered in this analysis. As future pandemics may involve different transmission mechanisms, it is crucial to develop models that incorporate both aerosol and fomite transmission to offer a more comprehensive risk assessment. Fourth, the current model does not account for social distancing between passengers, nor does it consider other mechanisms such as viral inactivation or deposition. These factors can be integrated into future studies. Fifth, the heterogeneity of infected populations and their travel patterns is not considered. In reality, transmission probabilities vary across different demographic groups, and travel patterns such as trip frequency or trip chains can result in superspreader events. Future research should account for demographic and travel behavior differences, as well as vaccination rates, to improve the accuracy of transmission risk predictions and policy evaluations.

Finally, the model focuses solely on transmission risks within train cars and does not consider risks at platforms, which may have poor ventilation. Future studies should extend the model to evaluate transmission risks throughout the entire journey, including time spent on platforms.

CRedit authorship contribution statement

Jiali Zhou: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Haris N. Koutsopoulos:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Because of confidentiality agreements, supporting data are not available from the authors.

Data availability

The authors do not have permission to share data.

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